

# Uchhas Saha

Dhaka, Bangladesh

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## SUMMARY

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Computer Science and Engineering undergraduate with experience in Software Engineering, Machine Learning, and DevOps fundamentals. Skilled in building scalable full-stack applications and deploying machine learning solutions using Flask, Django, Spring Boot, ReactJS, and SQL. Strong problem-solving ability with leadership experience in team-based and cross-campus initiatives. Interested in Software Engineering, Machine Learning Engineering, and DevOps roles.

## EXPERIENCE

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### Machine Learning Intern

CodeAlpha

Apr 2025 – Jun 2025

- Developed and optimized machine learning models for prediction and classification tasks.
- Applied feature engineering, model evaluation, and performance tuning techniques.
- Built structured data pipelines for machine learning workflows.
- Worked with NLP libraries like Spacy and NLTK.

### Campus Director

G17 Global

Jun 2024 – Apr 2025

- Led student teams to organize campus campaigns and events.
- Managed cross-campus collaboration and stakeholder communication.
- Coordinated execution of sustainability and awareness initiatives.

### Junior Executive (Finance & Marketing)

Robotics Club, BRAC University

Feb 2024 – Aug 2024

- Conducted financial analysis and project feasibility evaluation.
- Supported marketing, branding, and event execution activities.

## PROJECTS

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### Pet Segmentation & Breed Classification – Oxford-IIIT Pet Dataset

- Implemented and benchmarked four deep learning architectures (U-Net, Attention U-Net, DenseNet, MobileNet) for semantic segmentation of pet images from the Oxford-IIIT Pet Dataset.
- Best-performing model, Attention U-Net, achieved a Test mIoU of 0.8866, Dice coefficient of 0.9398, and Pixel Accuracy of 0.9414 across train, validation, and test splits.
- Integrated multi-label classification to predict pet species (cat/dog) and breed alongside segmentation masks.
- Applied data augmentation strategies to improve generalization across 37 pet categories.

### News Category Classifier – Multi-Class NLP Text Classification

- Conducted a comparative study of 24 model-preprocessing-representation combinations to classify news into four categories: Business, Science & Technology, World News, and Sports.
- Experimented with TF-IDF and Skip-gram representations across three preprocessing strategies (none, optimum, extreme) using ML models (Logistic Regression, Naive Bayes, Random Forest) and seven neural architectures (DNN, SimpleRNN, GRU, LSTM, and their bidirectional variants).

- Best result achieved by GRU with optimum preprocessing and Skip-gram embeddings: Accuracy 92.07%, Macro F1 0.9207, on a balanced 12,000-sample test set.
- Addressed training class imbalance (93,151 samples; Business-dominant at 39%) using macro F1 as the primary evaluation metric; conducted thorough EDA and manual hyperparameter tuning throughout.

#### **Banijjo – Stock Market Simulation Platform**

- Developed full-stack application using Spring Boot, ReactJS, REST APIs, and MySQL.
- Designed scalable backend services and integrated real-time data handling.

#### **Aureate Adhyay – Bookstore Management System**

- Built full-stack system using Flask and MySQL with authentication, CRUD operations and Flask Blueprint.
- Designed relational database schema and backend architecture.

#### **Credit Card Fraud Detection – Imbalanced Binary Classification**

- Built a fraud detection pipeline on a real-world credit card transaction dataset exhibiting severe class imbalance, addressed using SMOTE oversampling prior to model training.
- Preprocessed data by removing PII and irrelevant columns, applying Label Encoding for categorical features, and standardizing numerical features with StandardScaler.
- Benchmarked three classifiers — Logistic Regression, Decision Tree, and Random Forest (100 estimators) — evaluated using AUC-ROC and Precision-Recall AUC curves; Random Forest achieved superior performance and was selected as the final model.
- Serialized the trained model, scaler, and label encoders via Pickle and implemented a production-ready `predict_fraud()` inference function for real-time transaction classification.

#### **Customer Churn Prediction – Banking Client Retention Model**

- Developed a binary churn prediction system on a banking dataset with 10 customer features including credit score, geography, age, balance, and account activity status.
- Designed a modular sklearn Pipeline integrating StandardScaler for numerical features and OneHotEncoder for categorical features (Geography, Gender), ensuring leak-free preprocessing across train and test splits.
- Compared Logistic Regression, Random Forest, and Gradient Boosting classifiers using AUC-ROC and classification reports; Gradient Boosting was selected as the best-performing model and persisted via Joblib.
- Implemented a reusable `predict_churn()` inference function enabling end-to-end churn prediction for new customer records from raw input data.

#### **Hostage or Not – 3D FPS Game**

- Developed a 3D First Person Shooting game using Python and OpenGL with rendering and physics logic.

#### **Software Quality Predictor**

- Built machine learning model for software quality prediction.
- Applied classification techniques and evaluation metrics.

## **EDUCATION**

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### **BRAC University**

BSc in Computer Science and Engineering

2022 – Present

### **Dhaka Residential Model College**

Higher Secondary Certificate (Science)

2018 – 2021

### **Mohammadpur Preparatory School**

Secondary School Certificate

2013 – 2018

## ACHIEVEMENTS

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- 2nd Runner-Up – Hult Prize On-Campus Program (2025)
- 1st Runner-Up – EWU National Robofest (2024)
- 2nd Runner-Up – TechSpectra LFR competition (2022)
- Champion – Artsy Overlap National Art Competition (2021)

## VOLUNTEERING

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### Campus Ambassador

G17 Global, 2024

Led student engagement campaigns and awareness programs.

## SKILLS

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**Programming Languages:** Python, Java, JavaScript, SQL, HTML/CSS

**Frameworks:** Flask, Django, Spring Boot, ReactJS

**Tools & Technologies:** Git, GitHub, MySQL, MS SQL Server, Jupyter, Docker, Kubernetes

**Core Concepts:** Software Engineering, Machine Learning, REST APIs, Object-Oriented Programming, Database Design, DevOps Fundamentals

## ADDITIONAL

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**Certifications:** HackerRank — Software Engineering Intern, Software Engineer, SQL (Intermediate), SQL (Basic), Problem Solving (Intermediate), Problem Solving (Basic)

**Languages:** Bengali (Native), English (Fluent)